

Lecture 12A Live Problems

An introduction to equilibrium

Points toward lab manual pages: (Lab 2, Section 6 plan is blank; Lab 3, Section 1 works LO1–LO4) (section L2 S6 / L3 S1)

1. Heparin. Let H be the amount of heparin in the bloodstream, with $\frac{dH}{dt} = 2 - 0.3H$.

- (a) What happens if $H = 0$? What happens if $H = 10$?
- (b) What is the equilibrium?

2. Crayfish. Let C be the number of crayfish in our streams, with $\frac{dC}{dt} = 0.04C - 200$.

- (a) What is the equilibrium point?
- (b) What happens when C is above or below this equilibrium?

3. Heparin dosing with impaired kidneys. A patient has impaired kidney function and metabolizes heparin at half the usual rate. We want to provide a continuous dose m , in mg/hour, by IV:

$$\frac{dH}{dt} = m - 0.15 H \quad (\text{instead of } m - 0.3H).$$

- (a) What is the equilibrium value of H , as a function of m ?
- (b) What dose (value of m) gives this patient the same equilibrium amount of heparin, 6.67 mg?

4. Trapping crayfish. The crayfish population is estimated at 20,000. Their reproduction yields a 4% increase every month. Let k be the number we trap and kill each month: $\frac{dC}{dt} = 0.04 C - k$.

- (a) What is the equilibrium value of C , as a function of k ?
- (b) What value of k suffices to drive the crayfish to extinction?